

An interior modular zero of a weight-36 Hecke eigenform, and a degree-six modular-zero polynomial

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Abstract

We record a concrete, fully verified object: a weight-36 level-one Hecke cusp eigenform whose two zeros in the upper half-plane form a complex-conjugate pair in the *interior* of the standard fundamental domain—off the arc $|\tau| = 1$ and off every reflection line. The six zeros of the three eigenforms of S_{36} are exactly the roots of a single irreducible degree-six polynomial $P \in \mathbb{Z}[X]$, obtained by eliminating the Hecke eigenvalue from the per-form “zero-quadratic.” All displayed identities are exact and machine-checkable. We ask whether this packaging is known.

1 Setup

Let $S_{36} = S_{36}(\mathrm{SL}_2(\mathbb{Z}))$, of dimension three, and let $j = E_4^3/\Delta$ be the modular invariant ($j(i) = 1728$, $j(\rho) = 0$, $\rho = e^{2\pi i/3}$). Since $M_{24} = \langle E_4^6, E_4^3\Delta, \Delta^2 \rangle = \Delta^2 \langle j^2, j, 1 \rangle$ and $S_{36} = \Delta \cdot M_{24}$, every $f \in S_{36}$ has the shape

$$f = \Delta^3 (x_1 j^2 + x_2 j + x_3), \quad (x_1, x_2, x_3) \in \mathbb{C}^3. \quad (1)$$

As Δ is nonvanishing on $\mathbb{H}$ and f vanishes to order one at the cusp, the two zeros of f in $\mathbb{H}$ lie at the points τ for which $j(\tau)$ is a root of the *zero-quadratic* $x_1 X^2 + x_2 X + x_3$.

The Hecke operator T_2 acts on S_{36} . In the basis $b_1 = \Delta E_4^6$, $b_2 = \Delta^2 E_4^3$, $b_3 = \Delta^3$ (equivalently $j^2 \Delta^3$, $j \Delta^3$, Δ^3 ; leading q -orders 1, 2, 3) it is the integer matrix with characteristic polynomial

$$\chi(x) = x^3 - 139656 x^2 - 59208339456 x - 1467625047588864, \quad (2)$$

irreducible over $\mathbb{Q}$ with $\mathrm{disc}(\chi) > 0$; thus the Hecke field $K = \mathbb{Q}[x]/(\chi)$ is the totally real cubic field of S_{36} (LMFDB newform 1.36.a). For an eigenvalue θ (so $a_2 = \theta$) the eigenform’s coordinates in (1) may be taken to be the cofactor column of $T_2 - \theta I$,

$$\begin{aligned} x_1(\theta) &= \theta^2 - 138240\theta - 24774967296, \\ x_2(\theta) &= 34629120000\theta + 1502706401280000, \\ x_3(\theta) &= -17915904000000\theta - 8902341033984000000, \end{aligned} \quad (3)$$

polynomials of degree ≤ 2 in θ .

2 The modular-zero polynomial

Eliminating the eigenvalue collects all modular zeros into one polynomial.

Proposition 1. *With χ and x_i as in (2)–(3), set*

$$P(X) = \text{Res}_\theta(\chi(\theta), x_1(\theta)X^2 + x_2(\theta)X + x_3(\theta)) / (\text{leading coeff}).$$

Then $P \in \mathbb{Z}[X]$ is given by

$$\begin{aligned} P(X) = & X^6 + 135408 X^5 - 59796227520 X^4 + 1622049586606080 X^3 \\ & - 22503506634178560000 X^2 - 8581622846717952000000 X \\ & + 36680837244125184000000000, \end{aligned}$$

is irreducible over $\mathbb{Q}$, and has $\text{disc}(P) < 0$. Its six roots are exactly the j -invariants of the upper-half-plane zeros of the three Hecke eigenforms of S_{36} .

All assertions in Proposition 1 are exact and verified by symbolic computation (integer Hecke matrix, symbolic resultant, irreducibility test); see §5. Because P is irreducible, the six modular-zero j -values form a single Galois orbit of degree six, and $\text{disc}(P) < 0$ forces exactly one complex-conjugate pair among them. Numerically the roots are the four real values

$$-331296.33, \quad -1398.56, \quad 1139.16, \quad 167923.73,$$

(zeros on the boundary $\text{Re } \tau = \pm \frac{1}{2}$, the arc $|\tau| = 1$, and the imaginary axis, according as $j < 0$, $0 \leq j \leq 1728$, or $j > 1728$), together with the complex pair $14112.00 \pm 14652.78 i$.

3 The interior zero

The complex pair belongs to the eigenform of smallest Hecke eigenvalue, $a_2 \approx -26808.0076$ (the middle root of χ): for this real embedding the discriminant $x_2^2 - 4x_1x_3$ of the zero-quadratic is negative, so its j -roots are non-real. The corresponding zero in $\mathbb{H}$ is the conjugate pair

$$\tau_0 = -0.1323718666 \dots + 1.5748742500 \dots i, \quad -\overline{\tau_0},$$

with

$$j(\tau_0) = 14112.0038 \dots + 14652.7835 \dots i, \quad |\tau_0| = 1.58043 \dots > 1, \quad |\text{Re } \tau_0| < \frac{1}{2}.$$

Hence τ_0 lies in the *interior* of the standard fundamental domain. That $f(\tau_0) = 0$ is exact, by (1), since $j(\tau_0)$ is a root of the zero-quadratic.

For contrast: the weight-24 and weight-28 cusp eigenforms each carry a single *free* zero confined to the real- j locus (it recodes a_2 , via $j = 696 - a_2$ and $j = 936 - a_2$ respectively; weight 28 also has a zero forced to ρ by the E_4 -factor of M_{16}). Weight 36 is the first weight $\equiv 0 \pmod{12}$ at which a cusp eigenform has two genuinely free zeros, and here one Galois orbit escapes to the interior.

4 Questions

The construction is elementary, and zeros of modular forms are classically studied: the zeros of E_k lie on the arc $|\tau| = 1$ (Rankin–Swinnerton-Dyer), and zeros of Hecke eigenforms equidistribute on the modular surface with respect to hyperbolic measure as $k \rightarrow \infty$ (Rudnick, and subsequent work), so interior zeros are a low-weight phenomenon. My questions concern this specific packaging.

Question 1. Is $P(X) = \text{Res}_\theta(\chi, \text{zero-quadratic})$ —the “modular-zero resultant” of the Hecke basis—a named or previously studied object? Is there a closed form or generating description across the weights $k \equiv 0 \pmod{12}$?

Question 2. Does the sign data of the zero-quadratic discriminant $\delta(\theta) = x_2(\theta)^2 - 4x_1(\theta)x_3(\theta)$ across the real embeddings of the Hecke field characterize which eigenforms have interior zeros—i.e. is “ f has an interior zero” equivalent to “ δ is negative at the embedding $a_2(f)$ ”? Is weight 36 the smallest weight admitting an interior cusp-form zero?

Question 3. Does the interior point τ_0 , or the sextic field $\mathbb{Q}(j_0)$, carry any arithmetic structure? (It is not a CM point, since $j_0 \notin \mathbb{R}$.)

5 Verification

Every displayed identity is reproduced from scratch by a short `sympy/mpmath` script: it builds the integer T_2 matrix on S_{36} , recovers χ and the cofactor coordinates (3), forms P as the resultant and checks it against the stated coefficients, certifies irreducibility and $\text{disc}(P) < 0$, and locates τ_0 to high precision with an independent vanishing check. Available on request.